

## Akulon® S223-HG6

## PA66-GF30

30% Glass Reinforced, Heat Stabilized

| Properties                                   | Typical Data | Unit              | Test Method     |
|----------------------------------------------|--------------|-------------------|-----------------|
| <b>Rheological properties</b>                |              |                   |                 |
|                                              | dry / cond   |                   |                 |
| Molding shrinkage (parallel)                 | 0.2/*        | %                 | ISO 294-4       |
| Molding shrinkage (normal)                   | 1.1/*        | %                 | ISO 294-4       |
| <b>Mechanical properties</b>                 |              |                   |                 |
|                                              | dry / cond   |                   |                 |
| Tensile modulus                              | 9500/7500    | MPa               | ISO 527-1/-2    |
| Stress at break                              | 190/140      | MPa               | ISO 527-1/-2    |
| Strain at break                              | 3/5          | %                 | ISO 527-1/-2    |
| Flexural modulus                             | 8400/-       | MPa               | ISO 178         |
| Flexural strength                            | 250/-        | MPa               | ISO 178         |
| Charpy impact strength (+23°C)               | 80/100       | kJ/m <sup>2</sup> | ISO 179/1eU     |
| Charpy impact strength (-30°C)               | 70/70        | kJ/m <sup>2</sup> | ISO 179/1eU     |
| Charpy notched impact strength (+23°C)       | 12/20        | kJ/m <sup>2</sup> | ISO 179/1eA     |
| Charpy notched impact strength (-30°C)       | 10/10        | kJ/m <sup>2</sup> | ISO 179/1eA     |
| <b>Thermal properties</b>                    |              |                   |                 |
|                                              | dry / cond   |                   |                 |
| Melting temperature (10°C/min)               | 260/*        | °C                | ISO 11357-1/-3  |
| Temp. of deflection under load (1.80 MPa)    | 245/*        | °C                | ISO 75-1/-2     |
| Temp. of deflection under load (0.45 MPa)    | 260/*        | °C                | ISO 75-1/-2     |
| Coeff. of linear therm. expansion (parallel) | 0.2/*        | E-4/°C            | ISO 11359-1/-2  |
| Coeff. of linear therm. expansion (normal)   | 0.7/*        | E-4/°C            | ISO 11359-1/-2  |
| Burning Beh. at 1.5 mm nom. thicken.         | HB/*         | class             | IEC 60695-11-10 |
| Thickness tested                             | 1.5/*        | mm                | IEC 60695-11-10 |
| Burning Beh. at thickness h                  | HB/*         | class             | IEC 60695-11-10 |
| Thickness tested                             | 0.71/*       | mm                | IEC 60695-11-10 |
| <b>Electrical properties</b>                 |              |                   |                 |
|                                              | dry / cond   |                   |                 |
| Relative permittivity (100Hz)                | 3.8/10       | -                 | IEC 60250       |
| Relative permittivity (1 MHz)                | 3.5/4.1      | -                 | IEC 60250       |
| Dissipation factor (100 Hz)                  | 90/2800      | E-4               | IEC 60250       |
| Dissipation factor (1 MHz)                   | 160/800      | E-4               | IEC 60250       |

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| Properties                 | Typical Data | Unit              | Test Method    |
|----------------------------|--------------|-------------------|----------------|
| Volume resistivity         | 1E12/1E10    | Ohm*m             | IEC 60093      |
| Surface resistivity        | */1E13       | Ohm               | IEC 60093      |
| Electric strength          | 30/25        | kV/mm             | IEC 60243-1    |
| Comparative tracking index | 500/500      | -                 | IEC 60112      |
| Other properties           | dry / cond   |                   |                |
| Water absorption           | 6/*          | %                 | Sim. to ISO 62 |
| Humidity absorption        | 1.6/*        | %                 | Sim. to ISO 62 |
| Density                    | 1360/-       | kg/m <sup>3</sup> | ISO 1183       |

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